

PHYS 357: Honours Quantum Physics 1

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Abstract

Contents

1	Introduction	1
2	Stern-Gerlach Experiments	1
2.1	The classical picture	1
2.1.1	Angular momentum	1
2.1.2	Magnetic moment of a circulating charge	1
2.1.3	Gyromagnetic ratio	2
2.2	Spin	2
2.2.1	Intrinsic moment and the g-factor	2
2.2.2	Magnetons	3
2.2.3	Values for the electron, proton and neutron	3
2.3	The Stern-Gerlach apparatus	4
2.3.1	Force in an inhomogeneous field	4
2.3.2	Silver atoms	5
2.3.3	The classical prediction	5
2.4	Quantization of S_z	5
2.4.1	Ket notation	5
2.5	Sequential Stern-Gerlach experiments	6
2.5.1	Experiment 1: SGz then SGz	6
2.5.2	Experiment 2: SGz then SGx	6
2.5.3	Experiment 3: SGz, SGx, then SGz	6
3	Quantum States as Vectors	7
3.1	Superposition	7
3.2	Bras and the inner product	7
3.3	Probability amplitudes	7
3.4	Normalization	8
3.5	The state $ +\hat{x}\rangle$	9
3.6	Global phase	9
3.7	Completeness	9
3.8	Expectation value and uncertainty	10
3.9	The state $ +\hat{y}\rangle$	10
3.9.1	Experiment 5: measuring S_y	10
3.9.2	Fixing the relative phase	11
3.10	Generalization to n -level systems	12
3.11	Expectation value as an operator	13
4	Rotation of Basis States and Matrix Mechanics	13
4.1	States as column vectors	13
4.2	Operators as matrices	13
4.3	The rotation operator	14
A	Appendix	14

1 Introduction

Textbook used is JS Townsend, *A Modern Approach to Quantum Mechanics*, 2nd ed., University Science Books 2012. Taught by Bill Coish. Mondays, Wednesdays and Fridays from 9:35 to 10:25 in Steward Biology Building N2/2. Course grading is divided as follows:

- 10% problem sets (must be handwritten scans).
- 40% quizzes (on Wednesdays, best 8 of 12).
- 20% midterms (best of 2, October 2nd and November 6th).
- 30% final exam.

2 Stern-Gerlach Experiments

2.1 The classical picture

2.1.1 Angular momentum

Definition 1 (Orbital angular momentum). *The orbital angular momentum of a particle is*

$$\vec{L} = \vec{r} \times \vec{p}, \quad (1)$$

where $\vec{r}$ is the position vector and $\vec{p} = m\vec{v}$ is the linear momentum.

For motion confined to the x - y plane, only the z component survives. On a circular path of radius r with speed v ,

$$L_z = mvr.$$

Definition 2 (Angular momentum of a rigid body). *For a rigid body¹ rotating about a fixed axis this is*

$$L = I_0\omega, \quad (2)$$

with I_0 the moment of inertia and ω the angular velocity.

Remark 1. *The moment of inertia I_0 measures how mass is distributed relative to the rotation axis.*

2.1.2 Magnetic moment of a circulating charge

Definition 3 (Magnetic moment). *A magnetic moment is a measure of the net circulation of electric charge. It is a vector quantity that determines the torque a magnet will experience in an external magnetic field.*

¹A rigid body is one that does not deform under the action of external forces.

The magnetic moment of a point particle of mass m and charge q in a circular orbit of radius r and period T is

$$\mu = \frac{IA}{c} = \frac{1}{c} \left(\frac{qv}{2\pi r} \right) (\pi r^2) = \frac{q}{2mc} mvr = \frac{q}{2mc} L_z, \quad (3)$$

where I is the current and A is the area enclosed by the orbit. A single circulating charge counts as a steady current $I = q/T = qv/2\pi r$ because it passes any given point on the loop once per period.

2.1.3 Gyromagnetic ratio

Definition 4 (Gyromagnetic ratio). *The gyromagnetic ratio γ is the constant of proportionality between the magnetic moment and the angular momentum,*

$$\vec{\mu} = \gamma \vec{L}. \quad (4)$$

For the circulating charge above,

$$\gamma = \frac{q}{2mc}. \quad (5)$$

Remark 2. *This is a purely classical result, since the charge is treated as a point particle on a definite orbit with a definite radius and speed, and nothing is quantized. It matters here because measuring $\vec{\mu}$ is how we measure $\vec{L}$, and it is what the Stern-Gerlach apparatus exploits.*

2.2 Spin

Definition 5 (Spin). *Spin $\vec{S}$ is an intrinsic angular momentum carried by a particle. It is a fixed property of the particle type, like charge or mass, and does not come from any motion through space. A stationary electron alone in empty space still has spin.*

In quantum mechanics the spin $\vec{S}$ replaces the orbital angular momentum $\vec{L}$ of the classical picture.

2.2.1 Intrinsic moment and the g-factor

The magnetic moment is still proportional to the angular momentum, but the gyromagnetic ratio picks up an extra dimensionless factor,

$$\vec{\mu} = \gamma \vec{S}, \quad \gamma = \frac{qg}{2mc}, \quad (6)$$

where g is the g-factor of the particle.

Definition 6 (g-factor). *The g-factor of a particle³ is the dimensionless number relating its magnetic moment to its spin angular momentum. It measures how much stronger or weaker the moment is than the classical value $\gamma = q/2mc$, which corresponds to $g = 1$.*

²Here c is in Gaussian units.

³Its value must be measured or derived from relativistic quantum theory.

2.2.2 Magnetons

Definition 7 (Planck constant). *The Planck constant h is the fundamental constant of quantum mechanics. It sets the scale at which quantization matters, and relates the energy of a photon to its frequency by $E = h\nu$. Its value is*

$$h \approx 6.626 \times 10^{-34} \text{ J}\cdot\text{s}. \quad (7)$$

It has units of $J \cdot s$, which is the same as angular momentum.

Definition 8 (Reduced Planck constant).

$$\hbar = \frac{h}{2\pi}. \quad (8)$$

It carries units of angular momentum, so $[\hbar] = [L_z] = \frac{\text{kg}\cdot\text{m}^2}{\text{s}}$.

Writing $\vec{S} = \hbar(\vec{S}/\hbar)$ with $\vec{S}/\hbar$ dimensionless, the $\hbar$ joins the gyromagnetic ratio. The combination $q\hbar/2mc$ then carries the units and sets the natural size of the moment at a given mass scale.

Definition 9 (Bohr magneton). *The Bohr magneton is the natural unit of magnetic moment for an electron,*

$$\mu_B = \frac{|e|\hbar}{2m_e c}. \quad (9)$$

Definition 10 (Nuclear magneton). *The nuclear magneton is the same combination with the proton mass, and is the natural unit for nucleons,*

$$\mu_N = \frac{|e|\hbar}{2m_p c}. \quad (10)$$

Since $m_p/m_e \approx 1836$, we have $\mu_N \approx \mu_B/1836$.

Definition 11 (Nucleons). *The subatomic particles that make up the nucleus of an atom, specifically protons and neutrons.*

The electron moment is then written as

$$\vec{\mu}_e = g_e \mu_B \frac{\vec{S}}{\hbar}, \quad (11)$$

and similarly for the proton and neutron with μ_N in place of μ_B .

2.2.3 Values for the electron, proton and neutron

Particle	Charge q	Mass m	g -factor	Gyromagnetic ratio γ
Electron	$- e $	m_e	$g_e \approx 2$	$g_e \mu_B / \hbar$
Proton	$+ e $	m_p	$g_p \approx 5.59$	$g_p \mu_N / \hbar$
Neutron	0	m_p	$g_n \approx -3.83$	$g_n \mu_N / \hbar$

Table 1: Charges, masses, g -factors and gyromagnetic ratios.

Remark 3. The classical derivation gives $\mu = (q/2mc)L_z$, which vanishes when $q = 0$. The neutron is neutral yet has a nonzero moment, so its $\vec{\mu}$ cannot come from a single circulating point charge. This is a first sign that the classical picture fails.

Remark 4. Since $m_p \gg m_e$, we get $\mu_N \ll \mu_B$ and therefore $|\vec{\mu}_p| \ll |\vec{\mu}_e|$. Stern-Gerlach deflection of nuclei is much weaker than for electrons.

2.3 The Stern-Gerlach apparatus

Definition 12 (Stern-Gerlach apparatus). A device that measures one component of a particle's magnetic moment. A beam of neutral atoms passes through a magnet shaped to produce a strong field gradient along a chosen axis, here z . The gradient deflects each atom by an amount proportional to μ_z , and a screen records where it lands. Since $\vec{\mu} = \gamma\vec{S}$, the position on the screen is a direct measurement of S_z .

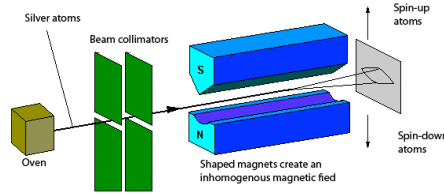


Figure 1: Illustration of the Stern-Gerlach apparatus.

2.3.1 Force in an inhomogeneous field

A magnetic moment in a field has potential energy

$$U = -\vec{\mu} \cdot \vec{B}, \quad (12)$$

so the force on it is

$$\vec{F} = -\vec{\nabla}U = \vec{\nabla}(\vec{\mu} \cdot \vec{B}), \quad F_z = -\frac{\partial U}{\partial z} = \mu_z \frac{\partial B_z}{\partial z}. \quad (13)$$

Definition 13 (Deflection). The transverse displacement of an atom on the screen, produced by the force F_z acting over the length of the magnet. It is proportional to μ_z , so the pattern on the screen is a record of the moments in the beam.

Remark 5. A uniform field gives no force, only a torque. The apparatus needs a field gradient to deflect anything.

2.3.2 Silver atoms

Silver is used because its net orbital angular momentum vanishes, $\vec{L} = 0$, while $\vec{S} \neq 0$. The entire magnetic moment comes from a single unpaired valence electron, so the atom behaves as one electron spin. For the electron,

$$\mu_z = -\frac{g_e |e|}{2m_e c} S_z, \quad (14)$$

and therefore

$$F_z = \mu_z \frac{\partial B_z}{\partial z} = -g_e \frac{|e|}{2m_e c} S_z \frac{\partial B_z}{\partial z} = g_e \frac{|e|}{2m_e c} \left| \frac{\partial B_z}{\partial z} \right| S_z, \quad (15)$$

where the last equality assumes $\partial B_z / \partial z < 0$.

Result 1.

$$F_z \propto S_z. \quad (16)$$

The deflection of an atom measures the z component of its spin directly.

2.3.3 The classical prediction

Spin is a vector,

$$\vec{S} = (S_x, S_y, S_z)^T, \quad S_z = |\vec{S}| \cos \theta. \quad (17)$$

Remark 6. *The spins leaving the oven point in random directions, so θ takes every value in $[0, \pi]$ across the beam and $S_z = |\vec{S}| \cos \theta$ takes every value in $[-|\vec{S}|, |\vec{S}|]$. Since $F_z \propto S_z$, each atom is deflected differently and the screen should show a continuous smear. It shows two spots.*

2.4 Quantization of S_z

Result 2 (Quantization of spin). *The electron spin angular momentum is quantized:*

$$S_z = \pm \frac{\hbar}{2}. \quad (18)$$

The beam splits into two spots.

2.4.1 Ket notation

Definition 14 (Ket notation). *A ket $|\cdot\rangle$ labels a quantum state as a vector in a vector space. For electron spin the two measurement outcomes give the basis states*

$$|\uparrow\rangle \text{ for } S_z = +\frac{\hbar}{2}, \quad |\downarrow\rangle \text{ for } S_z = -\frac{\hbar}{2}.$$

The space is two-dimensional, so any spin state is a linear combination of the two.

Remark 7. *From here on we follow Townsend and write $|+\hat{z}\rangle \equiv |\uparrow\rangle$ and $|-\hat{z}\rangle \equiv |\downarrow\rangle$. The same labels work for any axis: $|\pm\hat{x}\rangle$ are the states with $S_x = \pm\hbar/2$.*

2.5 Sequential Stern-Gerlach experiments

We write SG_z for a device whose gradient points along z , and SG_x for the same device rotated by 90° about the beam axis. Each device sends atoms into two beams. Blocking one of them turns the device into a filter: every atom that gets through has a known spin component.

A beam straight from the oven always splits 50/50 in any SG device, whatever its orientation. The interesting results come from sending a filtered beam into a second (or third) device.

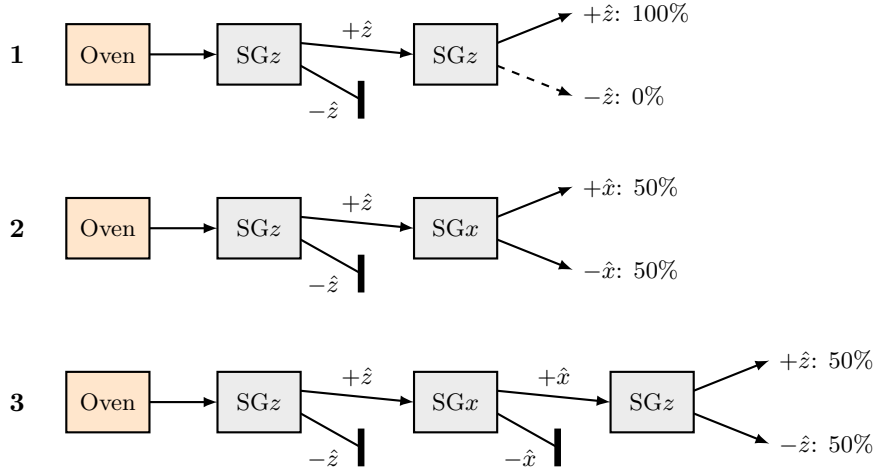


Figure 2: Sequential Stern-Gerlach experiments. Thick bars block a beam. Percentages are the fractions of atoms entering the last device.

2.5.1 Experiment 1: SG_z then SG_z

Filter the $+\hat{z}$ beam and send it into a second SG_z . Every atom comes out $+\hat{z}$, none come out $-\hat{z}$. The measurement is repeatable: once an atom is found with $S_z = +\hbar/2$, measuring S_z again gives the same answer. The atom leaves the first device in the state $|+\hat{z}\rangle$.

2.5.2 Experiment 2: SG_z then SG_x

Filter the $+\hat{z}$ beam and send it into an SG_x . Half the atoms come out $+\hat{x}$ and half come out $-\hat{x}$. Knowing $S_z = +\hbar/2$ tells us nothing about S_x .

2.5.3 Experiment 3: SG_z , SG_x , then SG_z

Filter $+\hat{z}$, then filter $+\hat{x}$, then measure S_z again. The beam splits 50/50 into $+\hat{z}$ and $-\hat{z}$, even though every atom was filtered to $+\hat{z}$ at the start. The S_x measurement wiped out the earlier S_z information.

September 11,
2026

Result 3. *An atom with a definite value of S_x does not have a definite value of S_z . Measuring S_x destroys what we knew about S_z , so no state has definite values of both at once.*

3 Quantum States as Vectors

3.1 Superposition

Experiment 3 shows that an atom in the state $|+\hat{x}\rangle$ does not have a definite $S_z = \pm\hbar/2$. It is neither $|+\hat{z}\rangle$ nor $|-\hat{z}\rangle$, but a combination of both.

Definition 15 (Superposition). *A general spin state is a linear combination of the basis kets,*

$$|\psi\rangle = c_+ |+\hat{z}\rangle + c_- |-\hat{z}\rangle, \quad c_{\pm} \in \mathbb{C}. \quad (19)$$

The pair $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$ is a basis, and the coefficients $c_{\pm}$ are complex numbers.

Remark 8. *This works exactly like an ordinary vector written in components,*

$$\vec{E} = E_x \hat{x} + E_y \hat{y} + E_z \hat{z}, \quad (20)$$

where the basis is orthonormal:

$$\hat{x} \cdot \hat{x} = \hat{y} \cdot \hat{y} = \hat{z} \cdot \hat{z} = 1, \quad \hat{x} \cdot \hat{y} = \hat{x} \cdot \hat{z} = \hat{y} \cdot \hat{z} = 0. \quad (21)$$

A component is picked out with a dot product, $E_x = \hat{x} \cdot \vec{E}$. Kets do the same thing, with the inner product playing the role of the dot product.

3.2 Bras and the inner product

Definition 16 (Bra). *Every ket $|\psi\rangle$ has a matching bra $\langle\psi|$. Putting a bra in front of a ket gives a complex number, the inner product*

$$\langle\phi| |\psi\rangle \equiv \langle\phi|\psi\rangle \in \mathbb{C}. \quad (22)$$

Definition 17 (Orthonormal basis). *The basis kets are orthonormal:*

$$\langle+\hat{z}|+\hat{z}\rangle = \langle-\hat{z}|-\hat{z}\rangle = 1, \quad \langle+\hat{z}|-\hat{z}\rangle = \langle-\hat{z}|+\hat{z}\rangle = 0. \quad (23)$$

3.3 Probability amplitudes

To pull out c_+ , act on $|\psi\rangle$ with $\langle+\hat{z}|$, just like dotting $\vec{E}$ with $\hat{x}$:

$$\langle+\hat{z}|\psi\rangle = c_+ \underbrace{\langle+\hat{z}|+\hat{z}\rangle}_{=1} + c_- \underbrace{\langle+\hat{z}|-\hat{z}\rangle}_{=0} = c_+. \quad (24)$$

The same steps with $\langle-\hat{z}|$ give c_- . So

$$c_{\pm} = \langle\pm\hat{z}|\psi\rangle. \quad (25)$$

Definition 18 (Probability amplitude). *The coefficient $c_{\pm} = \langle \pm \hat{z} | \psi \rangle$ is the probability amplitude for measuring $S_z = \pm \hbar/2$. The probability itself is the modulus squared,*

$$P(S_z = \pm \hbar/2) = |c_{\pm}|^2 = |\langle \pm \hat{z} | \psi \rangle|^2. \quad (26)$$

Principle 1 (Born rule). *The probability that a system in the state $|\psi\rangle$ is found in the state $|\phi\rangle$ is*

$$P(\phi | \psi) = |\langle \phi | \psi \rangle|^2 \in [0, 1]. \quad (27)$$

3.4 Normalization

A system in $|\psi\rangle$ is found in $|\psi\rangle$ with certainty, so

$$P(\psi | \psi) = |\langle \psi | \psi \rangle|^2 = 1. \quad (28)$$

This fixes what the bra looks like. Write it with unknown coefficients, $\langle \psi | = c'_+ \langle +\hat{z} | + c'_- \langle -\hat{z} |$. Then

$$\langle \psi | \psi \rangle = (c'_+ \langle +\hat{z} | + c'_- \langle -\hat{z} |) (c_+ | +\hat{z} \rangle + c_- | -\hat{z} \rangle) = c'_+ c_+ + c'_- c_-, \quad (29)$$

since the cross terms vanish by orthonormality. The two outcomes $S_z = \pm \hbar/2$ must have probabilities $|c_+|^2$ and $|c_-|^2$ that add to 1. For $\langle \psi | \psi \rangle$ to be that sum, each product $c'_\pm c_\pm$ must be real and equal to $|c_\pm|^2$. This forces $c'_\pm = c_\pm^*$.

Result 4. *Going from a ket to its bra, conjugate the coefficients:*

$$\langle \psi | = c_+^* \langle +\hat{z} | + c_-^* \langle -\hat{z} |. \quad (30)$$

In particular $\langle \psi | \pm \hat{z} \rangle = c_\pm^ = \langle \pm \hat{z} | \psi \rangle^*$. More generally, $\langle \phi | \psi \rangle = \langle \psi | \phi \rangle^*$.*

Definition 19 (Normalization condition).

$$\langle \psi | \psi \rangle = \langle \psi | +\hat{z} \rangle \langle +\hat{z} | \psi \rangle + \langle \psi | -\hat{z} \rangle \langle -\hat{z} | \psi \rangle = |c_+|^2 + |c_-|^2 = 1. \quad (31)$$

In words, $P(S_z = +\hbar/2) + P(S_z = -\hbar/2) = 1$: the atom is always found in one of the two beams.

Example. *Take*

$$|\psi\rangle = \frac{1}{\sqrt{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle.$$

The amplitude for $S_z = +\hbar/2$ is

$$\langle +\hat{z} | \psi \rangle = \frac{1}{\sqrt{3}} \underbrace{\langle +\hat{z} | +\hat{z} \rangle}_{=1} + i\sqrt{\frac{2}{3}} \underbrace{\langle +\hat{z} | -\hat{z} \rangle}_{=0} = \frac{1}{\sqrt{3}},$$

so $P(S_z = +\hbar/2) = 1/3$. In the same way $\langle -\hat{z} | \psi \rangle = i\sqrt{2/3}$, so $P(S_z = -\hbar/2) = |i|^2 \cdot 2/3 = 2/3$. The two add to 1, so the state is normalized. The factor i changes nothing here since $|i| = 1$.

3.5 The state $|+\hat{x}\rangle$

In Experiment 3, atoms in $|+\hat{x}\rangle$ enter an SGz and split 50/50:

$$P(S_z = \pm \frac{\hbar}{2}) = |\langle \pm \hat{z} | +\hat{x} \rangle|^2 = \frac{1}{2}. \quad (32)$$

So both amplitudes have modulus $1/\sqrt{2}$, and each can carry its own phase:

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} e^{i\delta_+} |+\hat{z}\rangle + \frac{1}{\sqrt{2}} e^{i\delta_-} |-\hat{z}\rangle = \frac{e^{i\delta_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad \delta = \delta_- - \delta_+, \quad (33)$$

with $\delta_{\pm} \in \mathbb{R}$.

3.6 Global phase

Take two kets that differ only by an overall phase, $|\psi\rangle = e^{i\alpha} |\psi'\rangle$ with $\alpha \in \mathbb{R}$. For any state $|\phi\rangle$,

$$P(\phi | \psi) = |\langle \phi | \psi \rangle|^2 = |e^{i\alpha} \langle \phi | \psi' \rangle|^2 = |\langle \phi | \psi' \rangle|^2 = P(\phi | \psi'), \quad (34)$$

since $|e^{i\alpha}| = 1$.

Result 5. $|\psi\rangle$ and $e^{i\alpha} |\psi\rangle$ give the same probabilities for every possible measurement, so they describe the same physical state. An overall (global) phase has no physical meaning and can be dropped.

Dropping the global phase $e^{i\delta_+}$ from $|+\hat{x}\rangle$ leaves

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle). \quad (35)$$

Remark 9. The relative phase δ is different: it sits between the two terms, so it cannot be pulled out front and dropped. Experiment 3 alone does not fix it. It is pinned down by combining more measurements (for example along y) with a phase convention, and the standard choice for $|+\hat{x}\rangle$ is $\delta = 0$.

3.7 Completeness

Remark 10. The Born rule, $P(\phi | \psi) = |\langle \phi | \psi \rangle|^2$, is the probability of finding $|\phi\rangle$ when $|\psi\rangle$ is prepared. Why is it true? There is no deeper derivation: it is a postulate that agrees with every experiment. As David Mermin put it, "shut up and calculate."

Proposition 1 (Completeness relation). Since any state can be expanded in the basis $\{|+\hat{z}\rangle, |-\hat{z}\rangle\}$, the sum of the projectors onto the basis states is the identity:

$$|+\hat{z}\rangle \langle +\hat{z}| + |-\hat{z}\rangle \langle -\hat{z}| = 1. \quad (36)$$

Applying this to $|+\hat{x}\rangle$ gives the expansion directly:

$$|+\hat{x}\rangle = (|+\hat{z}\rangle \langle +\hat{z}| + |-\hat{z}\rangle \langle -\hat{z}|) |+\hat{x}\rangle = \langle +\hat{z} | +\hat{x} \rangle |+\hat{z}\rangle + \langle -\hat{z} | +\hat{x} \rangle |-\hat{z}\rangle, \quad (37)$$

which is the general form of the previous subsection with $c_{\pm} = \langle \pm \hat{z} | +\hat{x} \rangle$.

3.8 Expectation value and uncertainty

Definition 20 (Expectation value). *The expectation value of S_z in the state $|\psi\rangle$ is the average outcome weighted by the probabilities,*

$$\langle S_z \rangle = \left(+\frac{\hbar}{2} \right) P(S_z = +\frac{\hbar}{2}) + \left(-\frac{\hbar}{2} \right) P(S_z = -\frac{\hbar}{2}) = \frac{\hbar}{2} |\langle +\hat{z}|\psi \rangle|^2 - \frac{\hbar}{2} |\langle -\hat{z}|\psi \rangle|^2. \quad (38)$$

Remark 11. *For N measurements on identically prepared atoms, with N_+ giving $+\hbar/2$ and N_- giving $-\hbar/2$, the measured average is*

$$\langle S_z \rangle_N = \frac{\hbar}{2} \frac{N_+ - N_-}{N}, \quad (39)$$

which tends to $\langle S_z \rangle$ as $N \rightarrow \infty$.

Definition 21 (Uncertainty). *The uncertainty ΔS_z is the standard deviation of the measurement outcomes,*

$$(\Delta S_z)^2 = \langle (S_z - \langle S_z \rangle)^2 \rangle = \langle S_z^2 \rangle - \langle S_z \rangle^2, \quad (40)$$

where $\langle S_z^2 \rangle$ is computed like $\langle S_z \rangle$ with the outcomes squared.

Example. *For $|+\hat{x}\rangle$, both probabilities are $1/2$, so*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{2} - \frac{\hbar}{2} \cdot \frac{1}{2} = 0, \quad \langle S_z^2 \rangle = \left(\frac{\hbar}{2} \right)^2 \cdot \frac{1}{2} + \left(-\frac{\hbar}{2} \right)^2 \cdot \frac{1}{2} = \frac{\hbar^2}{4},$$

and therefore $\Delta S_z = \hbar/2$. The average is zero but every single measurement gives $\pm\hbar/2$, so the spread is maximal.

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}}|+\hat{z}\rangle + i\sqrt{\frac{2}{3}}|-\hat{z}\rangle$, the probabilities are $1/3$ and $2/3$, so*

$$\langle S_z \rangle = \frac{\hbar}{2} \cdot \frac{1}{3} - \frac{\hbar}{2} \cdot \frac{2}{3} = -\frac{\hbar}{6}, \quad \langle S_z^2 \rangle = \frac{\hbar^2}{4} \left(\frac{1}{3} + \frac{2}{3} \right) = \frac{\hbar^2}{4}.$$

Then

$$(\Delta S_z)^2 = \frac{\hbar^2}{4} - \frac{\hbar^2}{36} = \frac{9\hbar^2 - \hbar^2}{36} = \frac{2}{9}\hbar^2, \quad \Delta S_z = \frac{\sqrt{2}}{3}\hbar \approx 0.47\hbar.$$

3.9 The state $|+\hat{y}\rangle$

3.9.1 Experiment 5: measuring S_y

Definition 22 (Cyclic permutation). *A cyclic permutation of the axes relabels them as $x \rightarrow y$, $y \rightarrow z$, $z \rightarrow x$ (or the reverse). It is a rotation of the coordinate system, so no physics changes under it.*

Take Experiment 3 (filter $+\hat{x}$, then measure S_z) and cyclically permute the axes so that $\text{SG}x$ becomes $\text{SG}z$ and $\text{SG}z$ becomes $\text{SG}y$. The result is Experiment 5: filter $+\hat{z}$, then measure S_y . By symmetry it must split 50/50 as well, so every cross-axis overlap has the same modulus:

$$|\langle +\hat{x} | +\hat{z} \rangle|^2 = |\langle +\hat{y} | +\hat{z} \rangle|^2 = |\langle -\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2}. \quad (41)$$

The same argument as for $|+\hat{x}\rangle$ then gives

$$|+\hat{y}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} |+\hat{z}\rangle + \frac{e^{i\gamma_-}}{\sqrt{2}} |-\hat{z}\rangle = \frac{e^{i\gamma_+}}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle), \quad \gamma = \gamma_- - \gamma_+. \quad (42)$$

3.9.2 Fixing the relative phase

Drop the global phases, $\gamma_+ = \delta_+ = 0$, so that

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\delta} |-\hat{z}\rangle), \quad |+\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + e^{i\gamma} |-\hat{z}\rangle). \quad (43)$$

The bra of $|+\hat{x}\rangle$ has conjugated coefficients, $\langle +\hat{x}| = \frac{1}{\sqrt{2}} (\langle +\hat{z}| + e^{-i\delta} \langle -\hat{z}|)$, so using orthonormality of the $\hat{z}$ basis,

$$\langle +\hat{x} | +\hat{y} \rangle = \frac{1}{2} (1 + e^{i(\gamma - \delta)}), \quad |\langle +\hat{x} | +\hat{y} \rangle|^2 = \frac{1}{2} (1 + \cos(\gamma - \delta)). \quad (44)$$

Requiring this to equal 1/2 forces $\cos(\gamma - \delta) = 0$, that is $\gamma - \delta = \pm\pi/2$. The remaining freedom is convention.

Result 6 (Phase convention). *Take $\delta = 0$ and $\gamma = +\pi/2$. Then*

$$|+\hat{x}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + |-\hat{z}\rangle), \quad |+\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle + i |-\hat{z}\rangle), \quad \langle +\hat{y}| = \frac{1}{\sqrt{2}} (\langle +\hat{z}| - i \langle -\hat{z}|). \quad (45)$$

The orthogonal partner is

$$|-\hat{y}\rangle = \frac{1}{\sqrt{2}} (|+\hat{z}\rangle - i |-\hat{z}\rangle), \quad (46)$$

which satisfies $\langle +\hat{y} | -\hat{y} \rangle = \frac{1}{2} (1 + (-i)(-i)) = \frac{1}{2} (1 - 1) = 0$.

Example. *For $|\psi\rangle = \sqrt{\frac{1}{3}} |+\hat{z}\rangle + i\sqrt{\frac{2}{3}} |-\hat{z}\rangle$, compute $\langle S_y \rangle$. The amplitudes are*

$$\begin{aligned} \langle +\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle +\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle +\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (-i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + \sqrt{\frac{2}{3}} \right), \\ \langle -\hat{y} | \psi \rangle &= \sqrt{\frac{1}{3}} \langle -\hat{y} | +\hat{z} \rangle + i\sqrt{\frac{2}{3}} \langle -\hat{y} | -\hat{z} \rangle = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} + i\sqrt{\frac{2}{3}} (i) \right) = \frac{1}{\sqrt{2}} \left(\sqrt{\frac{1}{3}} - \sqrt{\frac{2}{3}} \right). \end{aligned}$$

Squaring, with $2\sqrt{\frac{1}{3}}\sqrt{\frac{2}{3}} = \frac{2\sqrt{2}}{3}$,

$$P(S_y = +\frac{\hbar}{2}) = \frac{1}{2} + \frac{\sqrt{2}}{3}, \quad P(S_y = -\frac{\hbar}{2}) = \frac{1}{2} - \frac{\sqrt{2}}{3},$$

which add to 1 as they must. Then

$$\langle S_y \rangle = \frac{\hbar}{2} \left[\left(\frac{1}{2} + \frac{\sqrt{2}}{3} \right) - \left(\frac{1}{2} - \frac{\sqrt{2}}{3} \right) \right] = \frac{\sqrt{2}}{3} \hbar \approx 0.47\hbar.$$

The factor i that did nothing for S_z matters here: it is a relative phase, and it is exactly what tilts the spin toward $+\hat{y}$.

3.10 Generalization to n -level systems

September 16,
2026

A spin- $\frac{1}{2}$ particle is a two-level system, and each of $\{|\pm\hat{x}\rangle\}$, $\{|\pm\hat{y}\rangle\}$, $\{|\pm\hat{z}\rangle\}$ is a basis. An n -level system has a basis $\{|j\rangle\}$, $j = 1, \dots, n$.

Definition 23 (Orthonormal and complete basis). *The basis is orthonormal if*

$$\langle i | j \rangle = \delta_{ij}, \quad (47)$$

and complete if every state can be expanded in it,

$$|\psi\rangle = \sum_j c_j |j\rangle, \quad c_j = \langle j | \psi \rangle, \quad \langle \psi | = \sum_j c_j^* \langle j|. \quad (48)$$

Completeness is equivalent to

$$\mathbb{K} = \sum_{j=1}^n |j\rangle \langle j|. \quad (49)$$

Example. For $n = 2$, $\mathbb{K} = |+\hat{n}\rangle \langle +\hat{n}| + |-\hat{n}\rangle \langle -\hat{n}|$ for any $\hat{n} \in \{\hat{x}, \hat{y}, \hat{z}\}$, so

$$|\psi\rangle = \langle +\hat{x} | \psi \rangle |+\hat{x}\rangle + \langle -\hat{x} | \psi \rangle |-\hat{x}\rangle = \langle +\hat{y} | \psi \rangle |+\hat{y}\rangle + \langle -\hat{y} | \psi \rangle |-\hat{y}\rangle.$$

Definition 24 (Observable). *An observable A has outcomes a_j with states $|a_j\rangle$ forming a complete orthonormal set. For spin- $\frac{1}{2}$, A could be S_x , S_y or S_z , with outcomes $\pm\hbar/2$. By the Born rule,*

$$P(A = a_j) = |\langle a_j | \psi \rangle|^2. \quad (50)$$

Proposition 2. *Normalization $\langle \psi | \psi \rangle = 1$ is the statement that the probabilities sum to 1.*

Proof.

$$\langle \psi | \psi \rangle = \left(\sum_i c_i^* \langle a_i | \right) \left(\sum_j c_j |a_j\rangle \right) = \sum_{i,j} c_i^* c_j \delta_{ij} = \sum_j |c_j|^2 = \sum_j P(A = a_j) = 1.$$

Equivalently, inserting the identity, $\sum_j \langle \psi | a_j \rangle \langle a_j | \psi \rangle = \langle \psi | \left(\sum_j |a_j\rangle \langle a_j| \right) | \psi \rangle = \langle \psi | \psi \rangle$. $\square$

September 18,
2026

3.11 Expectation value as an operator

Proposition 3.

$$\langle A \rangle = \sum_j a_j |\langle a_j | \psi \rangle|^2 = \sum_j \langle \psi | a_j \rangle a_j \langle a_j | \psi \rangle = \langle \psi | \hat{A} | \psi \rangle, \quad \hat{A} = \sum_j a_j |a_j\rangle \langle a_j|. \quad (51)$$

4 Rotation of Basis States and Matrix Mechanics

4.1 States as column vectors

Definition 25 (Representation of a state). *In the S_z basis, a ket is represented by the column of its components and a bra by the conjugate row, the adjoint:*

$$|\psi\rangle \xrightarrow{S_z} \psi = \begin{pmatrix} c_+ \\ c_- \end{pmatrix}, \quad \langle\psi| \xrightarrow{S_z} \psi^\dagger = (c_+^* \quad c_-^*), \quad (52)$$

with $c_\pm = \langle \pm \hat{z} | \psi \rangle$. Then

$$\langle \psi | \psi \rangle = \psi^\dagger \psi = |c_+|^2 + |c_-|^2. \quad (53)$$

Remark 12. Write $|\psi\rangle \xrightarrow{S_z} \psi$, not $|\psi\rangle = \psi$. The ket is basis-independent; the column depends on the basis, just like $\vec{E}$ versus its components.

Example.

$$|\pm \hat{x}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad |\pm \hat{y}\rangle \xrightarrow{S_z} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}.$$

Example (Other bases). *In the S_x basis, $|\psi\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | \psi \rangle \\ \langle -\hat{x} | \psi \rangle \end{pmatrix}$, and similarly for S_y . So the same state has different columns:*

$$|+\hat{z}\rangle \xrightarrow{S_z} \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_x} \begin{pmatrix} \langle +\hat{x} | +\hat{z} \rangle \\ \langle -\hat{x} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |+\hat{z}\rangle \xrightarrow{S_y} \begin{pmatrix} \langle +\hat{y} | +\hat{z} \rangle \\ \langle -\hat{y} | +\hat{z} \rangle \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}.$$

4.2 Operators as matrices

September 21,
2026

Definition 26 (Projection operator). $P_{\pm \hat{n}} = |\pm \hat{n}\rangle \langle \pm \hat{n}|$, so that $\mathbb{K} = P_{+\hat{n}} + P_{-\hat{n}}$ and $P_{+\hat{z}} |\psi\rangle = c_+ |+\hat{z}\rangle$.

Definition 27 (Matrix of an operator). *In a basis $\{|j\rangle\}$, an operator A is represented by the matrix*

$$A_{ij} = \langle i | A | j \rangle. \quad (54)$$

Inserting the identity, $\langle k | A | \psi \rangle = \sum_j \langle k | A | j \rangle \langle j | \psi \rangle$, so $A | \psi \rangle$ is represented by the matrix product $A\psi$.

Example.

$$P_{+\hat{z}} \xrightarrow{S_z} \begin{pmatrix} \langle +\hat{z} | P_{+\hat{z}} | +\hat{z} \rangle & \langle +\hat{z} | P_{+\hat{z}} | -\hat{z} \rangle \\ \langle -\hat{z} | P_{+\hat{z}} | +\hat{z} \rangle & \langle -\hat{z} | P_{+\hat{z}} | -\hat{z} \rangle \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} c_+ \\ c_- \end{pmatrix} = \begin{pmatrix} c_+ \\ 0 \end{pmatrix}.$$

Definition 28 (Adjoint). Writing $A = \mathbb{K} A \mathbb{K} = \sum_{i,j} A_{ij} |i\rangle \langle j|$, the adjoint is

$$A^\dagger = \sum_{i,j} A_{ij}^* |j\rangle \langle i|, \quad (A^\dagger)_{ij} = A_{ji}^*. \quad (55)$$

If $|\psi'\rangle = A|\psi\rangle$, then $\langle\psi'| = \langle\psi| A^\dagger$.

4.3 The rotation operator

Definition 29 (Rotation operator). $R(\theta\hat{n})$ rotates a state by θ about $\hat{n}$: $|\psi'\rangle = R(\theta\hat{n})|\psi\rangle$. For example, $R(\frac{\pi}{2}\hat{y})$ takes $|\pm\hat{z}\rangle$ to $|\pm\hat{x}\rangle$ (up to phase), so

$$R\left(\frac{\pi}{2}\hat{y}\right)(c_+|+\hat{z}\rangle + c_-|-\hat{z}\rangle) = c_+|+\hat{x}\rangle + c_-|-\hat{x}\rangle. \quad (56)$$

Remark 13. The bra of $|\psi'\rangle$ is $\langle\psi|R^\dagger$, not $\langle\psi|R$. With the wrong choice, $|\psi\rangle = |+\hat{z}\rangle$ and $\theta = \pi/2$ would give $\langle\psi'|\psi'\rangle = \langle+\hat{z}|R(\pi\hat{y})|+\hat{z}\rangle = \langle+\hat{z}|-\hat{z}\rangle = 0$ instead of 1.

Theorem 1. Rotation operators are unitary:

$$R^\dagger R = R R^\dagger = \mathbb{K}. \quad (57)$$

Proof. A rotation must preserve normalization: $\langle\psi'|\psi'\rangle = \langle\psi|R^\dagger R|\psi\rangle = 1$ for every $|\psi\rangle$ and every $R(\theta\hat{n})$, which forces $R^\dagger R = \mathbb{K}$. $\square$

A Appendix

B Useful Links

Stern-Gerlach experiment: <https://www.youtube.com/shorts/QG7kJXxN1Kk>